Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Oblique Collisions

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A smooth sphere hits a fixed smooth vertical wall at 30° to the normal with speed u and $e = \frac{1}{2}$. Find the components of velocity after impact. **[4]**

2. Find the angle of rebound in the previous question. **[3]**

3. State which component of velocity is unchanged when a sphere hits a smooth wall. **[1]**

4. A particle hits a smooth plane at angle α to the plane with speed u. Coefficient e. Find the rebound angle to the plane. **[4]**

5. Two smooth spheres of equal mass collide. One is at rest. The line of centres makes angle θ with the velocity of the first. Given $e = 1$, show the paths after impact are perpendicular. **[5]**

6. Resolve velocities along and perpendicular to the line of centres for an oblique impact of two spheres. **[3]**

7. A sphere of mass $2m$ moving at u collides with a stationary sphere of mass m. Line of centres at 30° to $\mathbf{u}$, $e = \frac{1}{2}$. Find the component of velocity of the first sphere along the line of centres after impact. **[5]**

8. Explain why the tangential components of velocity are unchanged in a collision between smooth spheres. **[2]**

9. A ball is projected at $20\,\mathrm{m\,s^{-1}}$ at 45° to a smooth vertical wall $10\,\mathrm{m}$ away. It hits the wall. Given $e = 1$, find where it lands. **[5]**

10. Find the impulse on a smooth wall when a particle of mass m hits at angle α to the normal with speed u and coefficient e. **[3]**

11. Two equal smooth spheres collide with relative velocity along the line of centres of magnitude $2u$, $e = \frac{1}{3}$. Find the relative velocity of separation along the line of centres. **[3]**

12. A particle slides down a smooth incline and hits a smooth horizontal floor at the bottom. Discuss the impact. **[4]**

13. State the conditions for a collision between two spheres to be glancing. **[2]**

14. Find the loss in KE when a sphere hits a smooth fixed wall at angle α to the normal with speed u and coefficient e. **[4]**

15. A sphere moving at u at 40° to a smooth wall rebounds at 30° to the wall. Find e. **[4]**

Section B – Hard (Q16–25)

16. Two smooth spheres of masses m and $2m$ collide. Velocities before impact are $2u$ at 30° to the line of centres and u along the line of centres (opposite). $e = \frac{1}{2}$. Find the velocities after impact. **[7]**

17. A ball is thrown at a smooth vertical wall. Show that if $e = 1$ the path is symmetric in a certain sense. **[4]**

18. A particle hits a rough plane ($e = 0$, friction sufficient to prevent sliding). Describe the rebound. **[3]**

19. Equal smooth spheres: one at rest, impact parameter such that line of centres is at 45° to velocity, $e = 1$. Find the angle between the final velocities. **[5]**

20. A sphere hits a smooth floor at 50° to the horizontal with speed $10\,\mathrm{m\,s^{-1}}$ and $e = 0.6$. Find the velocity just after impact. **[5]**

21. Derive the condition that two equal smooth spheres moving perpendicular to each other before impact move perpendicular after impact when $e = 1$. **[5]**

22. A particle is projected from a point on a smooth inclined plane of angle α, hits the plane again. Discuss with e. **[5]**

23. Two spheres collide and the line of centres is parallel to the x-axis. Write the restitution equation for the x-components only. **[2]**

24. Find the direction of the impulse in a collision between two smooth spheres. **[2]**

25. A ball with $e = \frac{1}{2}$ is projected at 30° to the horizontal from ground level and hits a smooth vertical wall at the highest point. Find the horizontal distance from the wall to the landing point. **[6]**

End of Worksheet • *Show all working clearly.*